Assignment 4: Hardware and auctions

CS 356 UT Austin · Fall 2026 · Student handout

Every team starts with the same network. Hardware is how you make yours different: more capacity where your traffic is, a short way to the far side of the ring, your own link to a big neighbour, or links that come back sooner after a failure. There is no shop. Everything is sold in three auctions, for credits, and nobody knows the right prices in advance, so you set them by bidding. This handout covers the five kinds of hardware, how an auction works, what happens after you buy, and Auction 1's lots. It adds to the A4 rule book (the last section of the A4 handout, updated Sat Nov 7), which still holds. Give this handout to your AI agent with the rule book and `rules.json`. Every number is a key in `rules.json`, which may change until the game opens.

1 Credits

Credits (cr) are separate in each world: 100 cr when the game opens (`credits.at_open`) and 100 cr every Monday after that (`credits.per_monday`), carried over up to 200 cr (`credits.carry_max`); credits held in bids (Section 3) do not count toward that cap. They buy hardware and nothing else, they are not dollars, and they are not part of your score. Hardware bought in a world stays in that world.

2 The five kinds of hardware

Hardware adds to the network every AS starts with (Table 1). Each item is one *unit*: you buy it in one world and install it on your own AS there. This section says what each does, and Table 2, at its end, sums them up. The numbers are today's: the current values are in `rules.json`, under the keys in the tables, and may change until the game opens. All rates are real rates (the practice worlds run at a tenth of them).

Link	Capacity	Latency	`rules.json`
Crossing: hub to hub, through your pass	100 Mb/s	10 ms	`network.crossing`
IXP port: to a neighbour, at each IXP	200 Mb/s	1 ms	`network.ixp_port`
Tier-1 link, at each hub	40 Mb/s	32 ms	`network.tier1_link`
Hub to access switch	1000 Mb/s	6 ms	`network.hub_to_access`
Buffer: 12,500 bytes on every port of every switch			`buffer_bytes_per_port`
Program memory: 128 MiB for your switch program			`switch_program.memory_bytes`

Table 1: The network every AS starts with.



Second cable. A second cable through your pass, beside your crossing, so your crossing carries 200 Mb/s instead of 100. It is in the same pass, so it fails whenever your crossing fails: more capacity, same risk.



Link across the ring. A link from your hub in one of your towns to a new switch of yours at the IXP directly across the ring (X_i to X_{i+4}, the dotted lines on the map). The new switch has a 200 Mb/s port at the far IXP, so you can make agreements with the ASes there, just as at your own two IXPs. The *ordinary* kind carries 100 Mb/s in 22 ms; the *microwave* kind half as much, 10 ms faster. Without one, the far side is several crossings and IXPs away,

or two Tier-1 links (64 ms); Trading's target is 30 ms and Calls' 80 ms (`classes`). The link carries nothing until you have an agreement with an AS at the far IXP: its value is the deals it makes possible. It never fails in the failure schedule, but your link to an AS at the far IXP fails when that AS's port there fails, as at your own IXPs.

Private interconnect. Your own link to one neighbour at one of your IXPs, beside the IXP port: 200 Mb/s more between your hub and theirs in that town, for traffic between the two of you only. The neighbour must agree, on the website, before it is installed. Traffic flows over it only while you have an active agreement; the link stays when an agreement ends, so a new agreement uses it again. It fails when that IXP port fails.

Upgrades, four kinds, each a separate lot:

- *link capacity*: 100 Mb/s more on one hub-to-access link of yours, or on a link you bought. Not on your crossing (that is the second cable) or a Tier-1 link (its capacity cannot be bought);
- *buffer*: 12,500 bytes more on every port of one of your switches, which doubles its buffer;
- *faster port*: 200 Mb/s more on one of your IXP ports, or on the far port of a link across the ring;
- *program memory*: 128 MiB more (256 MiB in all) for your switch program on your switches in this world, not for its routing-policy calls. You submit one program for all four worlds, and it must still pass its dry run at 128 MiB, so the upgrade is headroom, not a new floor.

Repair crew. After a failure in this world, each of your links that failed comes back in half the time it would otherwise have taken: your crossing and second cable in a pass failure, your IXP ports and any private interconnect on them. A pass failure is shorter, not gone, so a backup agreement through the other pass can still pay.

Traffic over a private interconnect, or over the far port of a link across the ring, is metered and settled like traffic at an IXP. Two units in the same place add up (two second cables add 200 Mb/s); a second repair crew adds nothing. A lot limits how many units one team may win in it (Section 5); what you own is not limited.

Item	What it adds	Fails	key in `rules.json`: `hardware`....
Second cable	+100 Mb/s on your crossing	with your crossing	`second_cable`
Link across the ring, ordinary	100 Mb/s, 22 ms, to your switch at the far IXP (200 Mb/s port)	never	`cross_ring.ordinary`, `cross_ring.far_port_mbps`
Link across the ring, microwave	the same, at 50 Mb/s and 12 ms	never	`cross_ring.microwave`
Private interconnect	+200 Mb/s to one neighbour at one IXP	with that IXP port	`private_interconnect`
Upgrade: link capacity	+100 Mb/s on one link	with that link	`upgrades.link_capacity_mbps`
Upgrade: buffer	+12,500 bytes per port, one switch	—	`upgrades.buffer_bytes_per_port`
Upgrade: faster port	+200 Mb/s on one IXP port	with that port	`upgrades.port_mbps`
Upgrade: program memory	+128 MiB in this world	—	`upgrades.program_memory_bytes`
Repair crew	your outages are half as long	—	`repair_crew.outage_factor`

Table 2: The hardware at a glance.

3 How an auction works

Every item in Section 2 is sold in an *open* auction: everyone sees the winning bids as they come in, and a new bid must beat the ones already there.

What is for sale

- A *lot* is one kind of item from Section 2 in one world, for example the second cables in world 2. It has a number of identical units and a *reserve*, its lowest price.
- Every team has a seat in every world, so every team may bid on every lot. The AI seats do not bid in Auction 1. A lot is paid for from that world's credits (Section 1).
- Each bid is for one unit. A team may hold winning bids for at most the lot's per-team limit of units (Section 5), counting joint bids it leads or joins.

Bidding

- **Winning bids.** At any moment the highest bids on a lot, one for each unit for sale, are *winning*. The website shows every lot's winning bids, with each bidder's pseudonym, and its *minimum bid*.
- **The minimum bid.** While a unit has no bid, it is the reserve. Once every unit has a bid, it is the lowest winning bid raised by one step: the fraction `auctions.min_raise` of that bid, rounded to whole credits (halves up), and at least 1 cr. So after 10 cr the minimum is 11 cr; after 20 cr, 22 cr; after 34 cr, 37 cr. You may bid any whole number of credits at or above the minimum.
- **Outbid.** While a unit has no bid, a new bid takes it. Once every unit has a bid, a new bid takes the unit of the lowest winning bid, which is then *outbid*. Of two equal winning bids, the later one is outbid first.
- **Your credits are held.** When you bid, that many credits leave your credits in that world at once, so you can never bid more than you have. If you are outbid, they come back at once and you may bid again. A bid cannot be withdrawn.

The close

- Each lot closes on its own, at first at the auction's closing time (`schedule.auctions_close`).
- **Late bids push the close back,** by less each time. A bid in the last e minutes before a lot's close moves that lot's close e minutes later, and e halves. The first e is `auctions.extension_minutes`, so the extensions are 10, 5, 2.5 and 1.25 minutes; once e would be under a minute there are no more. Every lot therefore closes less than 20 minutes after the closing time.
- **At the close** each winning bid wins one unit, and the team pays its own bid: the credits already held. Units with no bid are not sold. Every bid and every outbid goes on the ledger as it happens.

How to bid

- **Decide your top price first**: what the item is worth to you (Section 2). Never bid above it.

	A bids 10 Mon	**B bids 12** Mon	**C bids 11** Tue	**A bids 13** Wed 12:00	**D bids 13** Wed 19:55	**B bids 14** Wed 20:08
unit 1	A 10	A 10	C 11	A 13	A 13	A 13
unit 2	no bid	B 12	B 12	B 12	D 13	B 14
minimum bid	10	11	12	13	14	14
lot closes	20:00	20:00	20:00	20:00	20:10	20:15
credits back			A: 10	C: 11	B: 12	D: 13

Figure 1: An example with made-up numbers: one world's lot of 2 second cables, reserve 10 cr, closing Wednesday at 20:00. Each column is one bid and the state just after it. C's 11 outbids A's 10, the lowest winning bid, and A's 10 cr come back; A then bids 13. D bids at 19:55, in the last 10 minutes, so the close moves to 20:10 and the next extension is 5 minutes. B's 14 at 20:08 outbids D (D's 13 is later than A's 13) and moves the close to 20:15. Nobody bids after that: A pays 13 cr and B 14 cr; C and D pay nothing.

- **Bid the minimum, and raise only when you are outbid.** You pay your own bid, so bidding more than you need to costs you the difference.
- **Your winning bids hold your credits.** Credits held on one lot cannot be spent on another, so you may not be able to win everything. Choose the items you need most.
- **You pay when you win,** even for an item that needs someone else: a link across the ring carries nothing until you have an agreement at the far IXP, and a private interconnect cannot be installed until the neighbour agrees. Line these up before you bid.
- **Watch the close.** Late bids near the closing time push the close back, so the last few minutes decide many lots.

Joint bids

- Several teams in the same world may bid together for one unit, with fixed shares of the cost, in whole credits, that add up to the bid.
- One team, the *lead*, places the bid and names its partners and their shares. The lead's credits hold the whole bid until each partner confirms on the website; then that partner's share is held from its own credits instead, and the lead gets it back. A share that is not confirmed, or is declined, stays on the lead, who pays it if the bid wins. If the bid is outbid, each team gets back what it put in.
- The lead holds what is won and installs it on its own AS, so agree with your partners where before you bid.
- A jointly bought private interconnect joins two partners' hubs. If either partner complains to us about how it is used, its capacity is split between their traffic in exactly the ratio each paid.

4 After you buy

- **Install.** What you win goes to your inventory in that world. Install it on your own AS there, choosing where (which link, switch or port, which site across the ring, which neighbour for a private interconnect). It works from the next cutoff. A private interconnect needs its

neighbour to agree first. Installed hardware appears to your switch program at start-up like the rest of your network; the API guide's Section 6 says where, and how to add it to a practice world to try it first.

- **Move.** You may move an item to another place in your AS; the move takes effect at the next cutoff.
- **Keep or sell.** Hardware is yours for the rest of the game. You may sell it to another team in the same world, for credits only (never for dollars or through an agreement), at any price you both agree on; the sale goes on the ledger, and the item goes to the buyer's inventory, not installed. Sales that only move credits can be reversed (rule book, sham deals).
- **What everyone sees.** The map on the website shows every world's installed hardware.

5 Auction 1

Auction 1 opens with the game, Mon Nov 9 at 18:00 (`schedule.game_opens`), and closes at the first entry of `schedule.auctions_close`, or a little later for a lot with late bids (Section 3). Hardware you win works from the next cutoff after you install it, so from Round 2 at the earliest. Its lots, the same in each world, are in `auctions.lots`: for each kind, the units for sale per world, the reserve and the most units one team may win. Auction 1 sells both kinds of link across the ring, the second cable, the repair crew, the private interconnect, and the buffer and program-memory upgrades. Lots for Auctions 2 and 3 are posted on the website the Monday before each one, and bidding opens when they are posted.